



American International University-Bangladesh (AIUB)

Faculty of Science and Technology (FST)
Department of Computer Science (CS)
Undergraduate Program

I. Course Code and Title:
COE 3206: Computer Networks

II. Credit:
3 credits

III. Nature:
Core Course for CS, CSE, CSSE, SE, CIS

Lab Manual

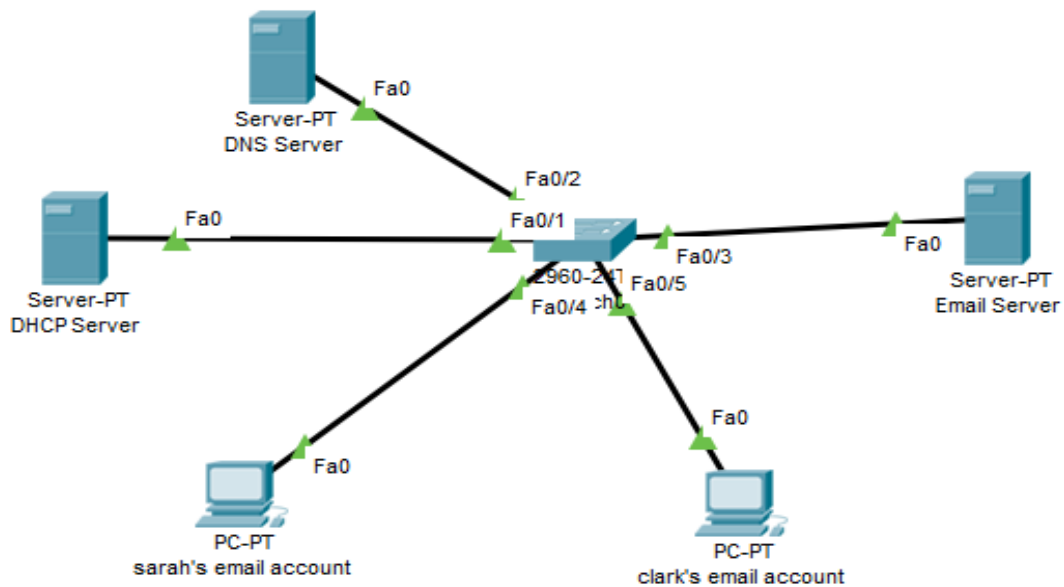
Title: Configuration of DNS, DHCP and Email servers

Abstract: This experiment is designed to introduce

1. DNS server configuration
2. DHCP server configuration
3. Email server configuration

Software: cisco packet tracer (version 7.3.0)

Network Design



1st server: DHCP server

Desktop tab

→ IP configuration: Configure it manually

- Configure IP address,
- Configure subnet mask and
- Configure DNS server

DHCP Server

Physical	Config	Services	Desktop	Programming	Attributes
☐ DHCP ☒ Static					
IP Address				10.0.0.1	
Subnet Mask				255.0.0.0	
Default Gateway					
DNS Server				10.0.0.2	

Service tab

Activate service by clicking 'on'

Assign 10.0.0.2 as DNS server

DHCP Server

Physical	Config	Services	Desktop	Programming	Attributes
SERVICES					
HTTP					
DHCP					
DHCPv6					
TFTP					
DNS					
SYSLOG					
AAA					

DHCP	
Interface	FastEthernet0
Pool Name	serverPool
Default Gateway	0.0.0.0
DNS Server	10.0.0.2

2nd Server: DNS server

Desktop tab

→ Ip configuration: Since this is a server, we will set a static IP configuration.

DNS Server

Physical	Config	Services	Desktop	Programming	Attributes
☐ DHCP					
☒ Static					
IP Address				10.0.0.2	
Subnet Mask				255.0.0.0	
Default Gateway				0.0.0.0	
DNS Server				10.0.0.2	

Service tab

Activate service by clicking 'on'

Write down 'gmail.com' in 'Name' box and '10.0.0.3' in the 'Address' box

Then, click "Add" to store the record.

 **DNS Server**

Physical Config **Services** Desktop Programming Attributes

SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL

DNS Service ☒ On

Resource Records

Name

Address

3rd Server: Email server

Desktop tab

→ IP configuration: Since this is a server, we will set a static IP configuration.

 **Email Server**

Physical Config Services **Desktop** Programming Attributes

☐ DHCP ☒ Static

IP Address

Subnet Mask

Default Gateway

DNS Server

Service tab

Write down 'gmail.com' in 'Domain name' box

Press 'set' button

Write down 'sarah' in 'user' box and '123' in 'password' box

Press '+' button

 Email Server

Physical Config **Services** Desktop Programming Attributes

SERVICES EM.

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL**

SMTP Service

☒ ON ☐ OFF

Domain Name:

User Setup

User Password

Write down 'clark' in 'user' box and '456' in 'password' box

Press '+' button

 Email Server

Physical Config **Services** Desktop Programming Attributes

SERVICES EMAIL

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL**
- FTP

SMTP Service

☒ ON ☐ OFF

Domain Name:

User Setup

User Password

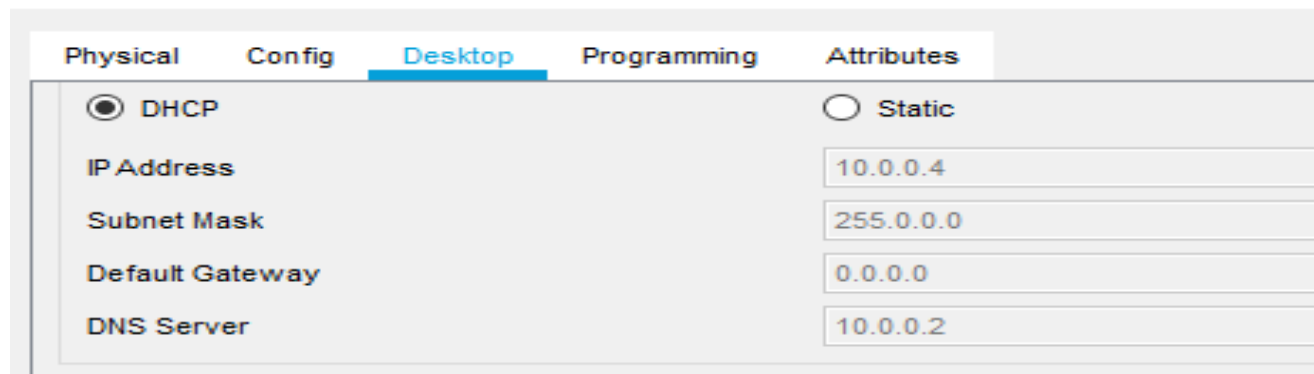
sarah

Pc0

Desktop tab

→ IP configuration: by clicking the DHCP option

 sarah's email account



Physical	Config	Desktop	Programming	Attributes
		☒ DHCP	☐ Static	
		IP Address	10.0.0.4	
		Subnet Mask	255.0.0.0	
		Default Gateway	0.0.0.0	
		DNS Server	10.0.0.2	

Desktop tab

→ Email tab: configure mail

Write down 'sarah' in 'Your name' box

Write down 'sarah@gmail.com' in 'Email address' box

Write down 'gmail.com' in 'Incoming mail server' box

Write down 'gmail.com' in 'Outgoing mail server' box

Write down 'sarah' in 'User Name' box

Write down '123' in 'password' box

Press 'save' button

sarah's email account

Physical Config **Desktop** Programming Attributes

Configure Mail

User Information

Your Name:

Email Address:

Server Information

Incoming Mail Server:

Outgoing Mail Server:

Logon Information

User Name:

Password:

Pc1

Desktop tab

→ IP configuration: by clicking the DHCP option

clark's email account

Physical Config **Desktop** Programming Attributes

☒ DHCP ☐ Static

IP Address:

Subnet Mask:

Default Gateway:

DNS Server:

IP Configuration

Desktop tab

→ Email tab: Configure mail

Write down 'clark' in 'Your name' box

Write down 'clark@gmail.com' in 'Email address' box

Write down 'gmail.com' in 'Incoming mail server' box

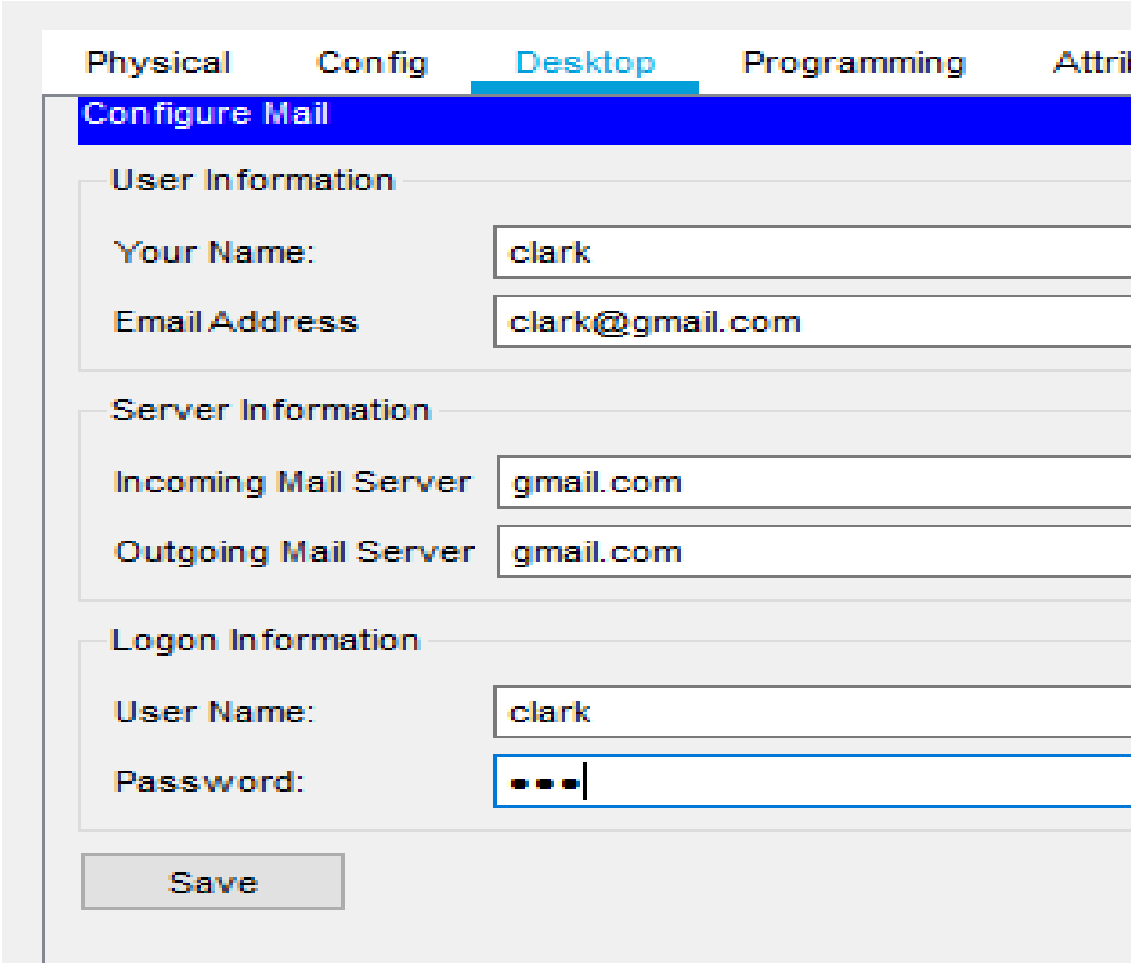
Write down 'gmail.com' in 'Outgoing mail server' box

Write down 'clark' in 'User Name' box

Write down '456' in 'password' box

Press 'save' button

clark's email account



Physical	Config	Desktop	Programming	Attril
Configure Mail				
User Information				
Your Name:		clark		
Email Address		clark@gmail.com		
Server Information				
Incoming Mail Server		gmail.com		
Outgoing Mail Server		gmail.com		
Logon Information				
User Name:		clark		
Password:		●●●		
Save				

Now you can compose and send mail

Open Desktop tab of PC0 and click Email. Then compose send email to Clark.

 sarah's email account

Physical Config **Desktop** Programming Attributes

Compose Mail

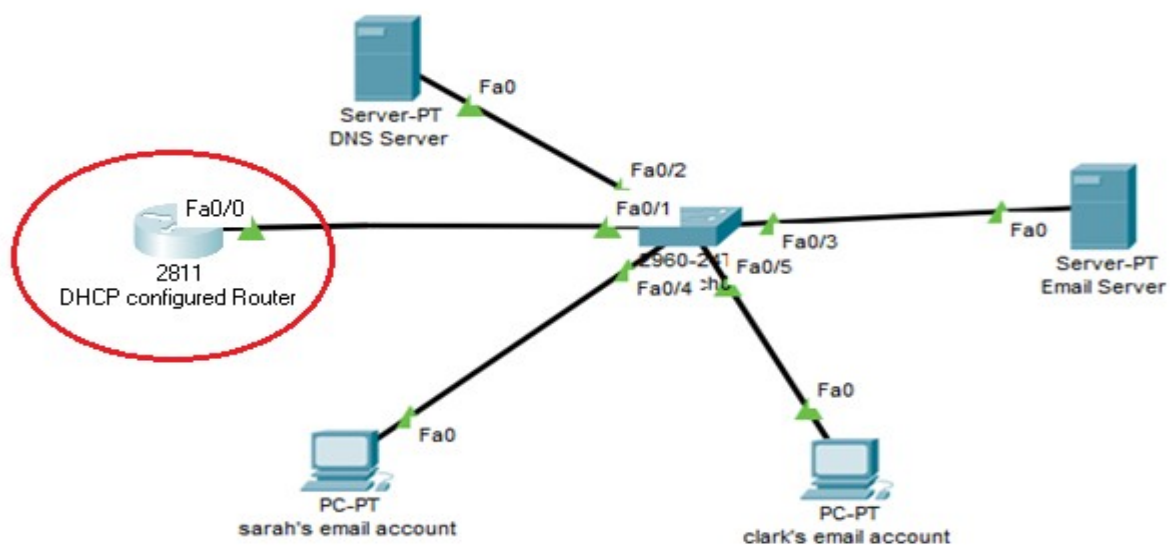
Send

To: clark@gmail.com

Subject: Hi

How are you?

You can configure DHCP in a router as well. Consider the following network,



Router configuration

```
Router>en
Router#conf t
Router(config)#int f0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#ip dhcp pool dpool
Router(dhcp-config)#network 10.0.0.0 255.0.0.0
Router(dhcp-config)#default-router 10.0.0.1
Router(dhcp-config)#dns-server 10.0.0.2
Router(dhcp-config)#exit
```

Note: All the other configuration remains the same.